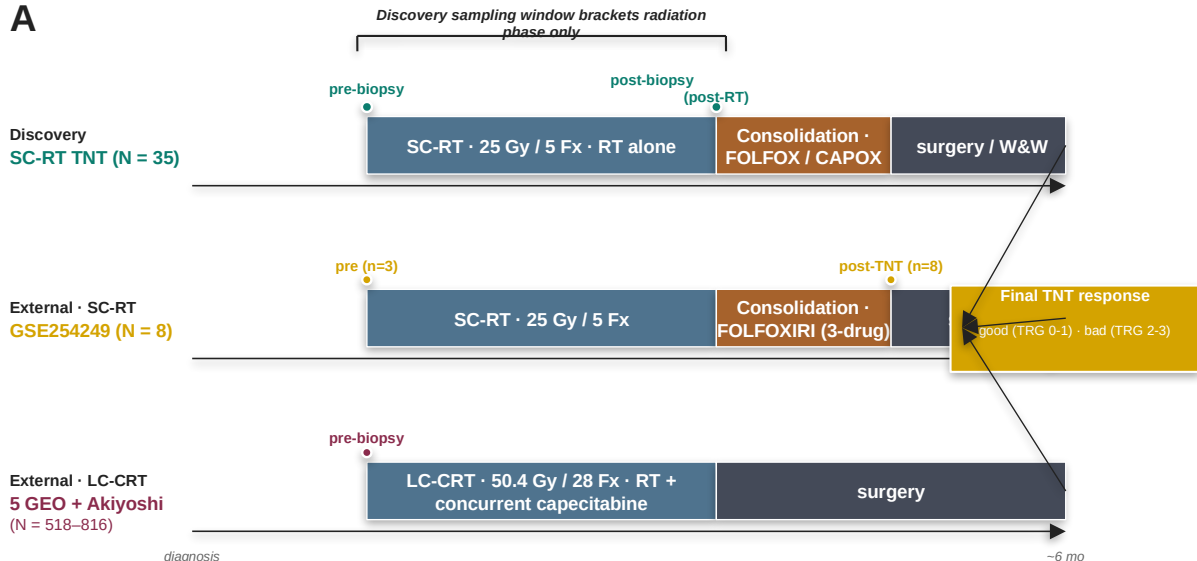
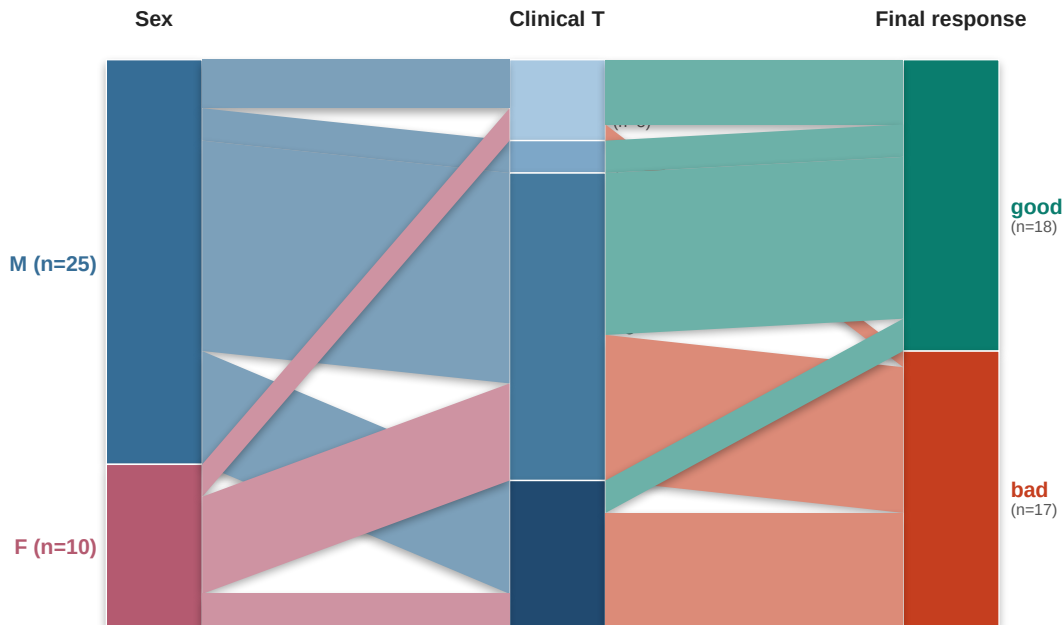


A



All three regimens converge to the same final-response endpoint. The pre-treatment baseline predictor (Fig 9) cannot encode downstream regimen choice and nevertheless reproduces across RT fractionation (SC-RT vs LC-CRT) and chemo-timing/backbone axes → regimen-agnostic.

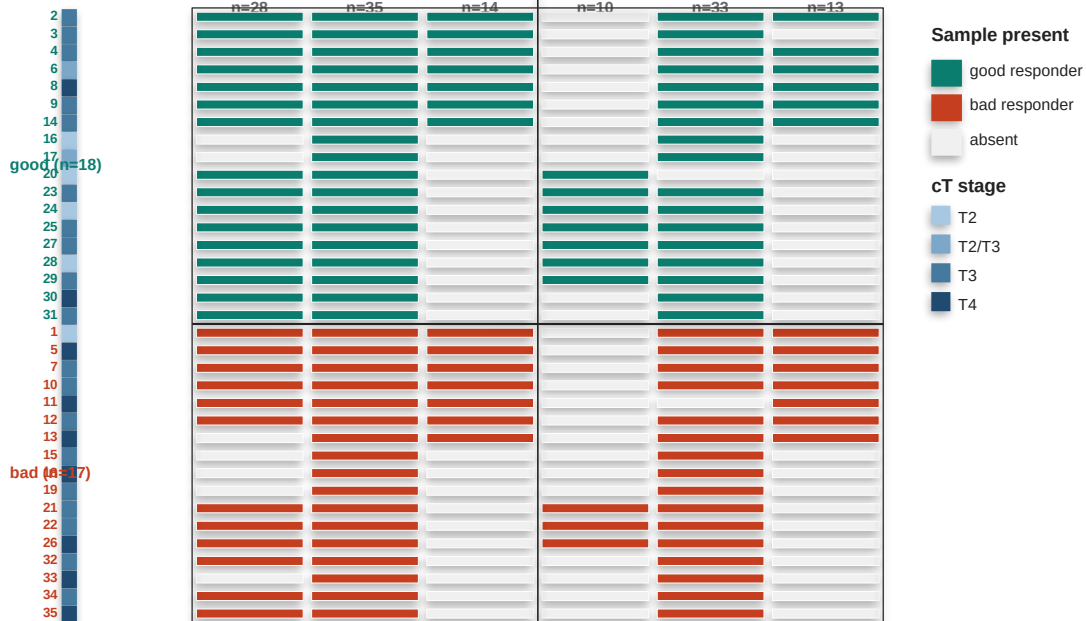
Paired analyses (Figs 6–8) are scoped to the discovery RT-phase window only.

B

N = 35 MSS LARC patients; ribbon widths proportional to subject counts.

T4 stage enriched in bad responders (7/9 vs 2/9 good, Fisher P = 0.086); no sig. sex association.

C



77 WES samples (28 normal + 35 pre + 14 post) · 56 RNA-seq samples (10 normal + 33 pre + 13 post)

12 subjects have paired pre + post RNA-seq (used in all paired-analysis figures).

D

DISCOVERY

Pre-CRT LASSO

AUC

4-feature ElasticNet,
nested outer-LOOCV

AUC (chance = 0.5)

0.745

[0.56, 0.90]

→ Fig 5

LC-CRT META

Thread 1 (5

cohorts, N = 518)

DSB Z=+3.17 · cell-cycle
+3.21 · E2F/MYC +2.79 ·
EMT +1.61 (trend)

Stouffer Z (|Z| = 1.96 = P 0.05)

+3.21

max Z

→ Fig 9A

LC-CRT META

Thread 2 (6

sources, N = 816)

CD8-cytotoxic + Akiyoshi
2023 (JAMA Netw Open)

Stouffer Z

+3.29

P = 0.001

→ Fig 9A

SC-RT EXTERNAL

GSE254249 (post-

TNT, N = 8)

7/7 signatures
concordant with
discovery · Tcell_infil MW
P = 0.036

signature direction concordance (7/7 = unanimous)

7/7

sign P = 0.016

→ Fig 9E

Radiation-phase pharmacodynami

CS
T good 6/6 up (P =
0.016) · IGH coherence
Wilcoxon P = 0.035

-log₁₀ P (horizontal bar length)

0.016

min P

→ Figs 6-7

Two externally-validated orthogonal pre-CRT axes (Thread 1 tumor-intrinsic + Thread 2 immune) reproduce across three regimen strata (SC-RT + FOLFOX/CAPOX; LC-CRT + concurrent cape; SC-RT + FOLFOXIRI), supporting a regimen-agnostic interpretation; *paired radiation-phase biopsies add an orthogonal dynamic layer (target engagement + directional immune coherence).*